125101 - Analytical chemistry 1 for engineers

Tutorial 11 – Electroanalysis

T11-1. The drug Librium gives a polarographic wave with $E_{1/2} = -0.265\ V$ (versus S.C.E.) in 0.05 M H_2SO_4. A 50.0mL sample containing Librium gave a wave height of 0.37μA. When 2.00 mL of 3.00 mM Librium in 0.05 M H_2SO_4 were added to the sample, the wave height increased to 0.8 μA. Find the molarity of Librium in the unknown.

T11-2. Ions that react with Ag^+ can be determined electrogravimetrically by deposition on a silver working anode:

$$Ag(s) + X^- \longrightarrow AgX(s) + e \qquad X = Cl^-, Br^-, I^-, SCN^-$$

a. What will be the final mass of a silver anode used to electrolyze 75.00 mL of 0.02380 M $KSCN$ (M.W. of $SCN-$ is 58.078 g/mol) if the initial mass of the anode is 12.4683g?

b. At what electrolysis voltage (versus S.C.E. with E = 0.241 V) will $AgBr(s)$ be deposited from 0.10 M Br^-?

$AgBr(s) + e \rightleftharpoons Ag(s) + Br^-$ with $E^o = 0.071V$ (Consider negligible current flow, so that there is no ohmic potential, concentration polarization, or over potential)

c. Is it theoretically possible to separate 99.99% of 0.1 M KI from 0.10 M KBr by controlled potential electrolysis?

$$Ksp(AgI) = 8.3 \times 10^{-17}$$

$$Ksp(AgBr) = 5.0 \times 10^{-13}$$

T11-3. A constant current of 0.800 A is used to deposit copper at the cathode and release oxygen at the anode of an electrolytic cell. Calculate the mass (in grams) of each product formed in 15.2 min. Assume that no other redox reaction occurs and use the following reactions:

$$Cu(aq)^{2+} + 2e \rightleftarrows Cu(s)$$

$$O_2(g) + 4H^+(aq) + 4e \rightleftarrows 2H_2O(l)$$

T11-4. The same set-up in **T11-3** was used for a **coulometric titration**. Several drops of Bromothymol blue indicator (pK_a = 7.1, pH range of 6.0 to 7.6 with the color change from yellow to blue) and 1.00 mL without knowing concentration KOH in the anode cell. Press the start button, to generate acid in the anode cell by circulating constant current of 0.800 A. When you observe color in the solution changing from yellow to blue, turn off the current and count the time of 15.2 min for the reactions. Try to compute the concentration of KOH.